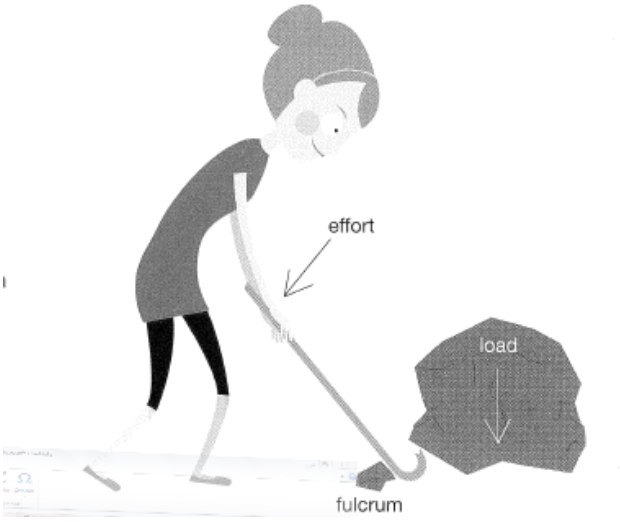
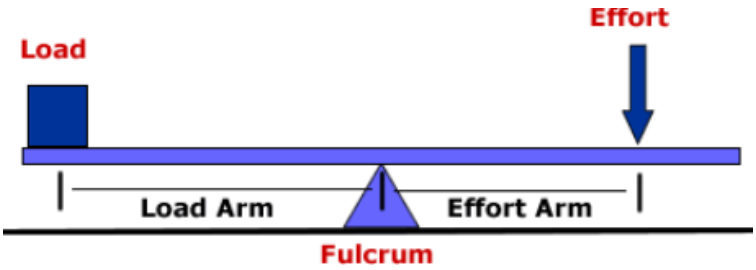
The lever

Read the text and fill in the definitions table below.

A **lever** is a simple machine that is made of a long, rigid object (such as a stick or metal rod) and a pivot or **fulcrum** about which it rotates. Most levers, like the crowbar shown in Figure 8.1.4, increase the size of the effort that you can apply or reduce the effort needed to get a job done.



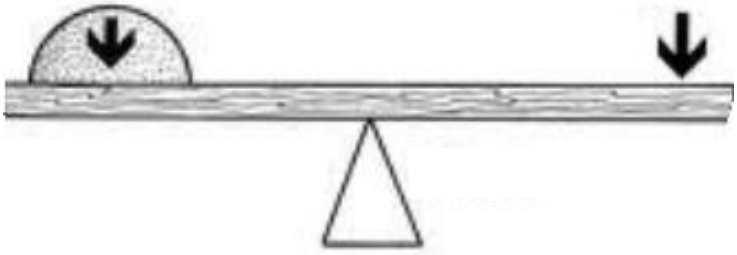
Levers can be represented using a diagram as shown below.
The load is the thing that needs to be lifted.
The effort is the force applied to move the object.
The load arm is the distance between the load and the fulcrum.
The effort arm is the distance between the effort and the fulcrum.



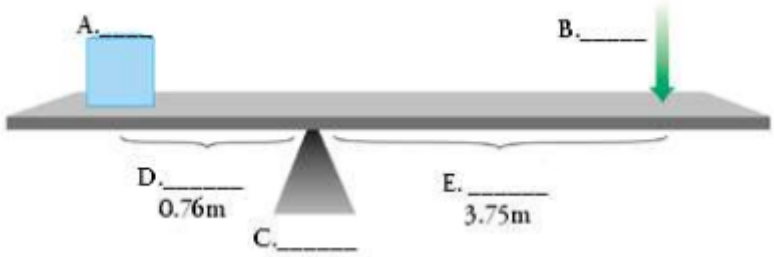
Term	Definition
	Point around which the lever rotates
	The object that needs to be lifted
	The force applied to lift the load
Effort arm	
Load arm	

Colour the lever diagram below using the following key.

Lever Feature	Colour
load	red
Effort arm	blue
Load arm	pink
Fulcrum	green

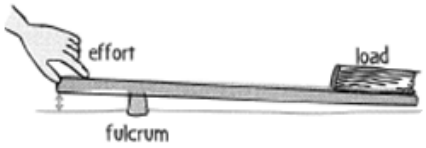
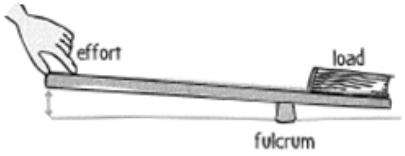
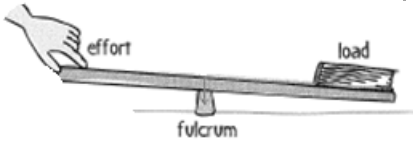


Label the lever diagram below.



Letter	Name of part	Letter	Name of part
A		D	
B		E	
C		F	

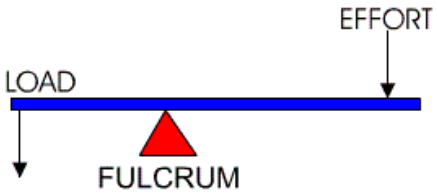
Into the table select the lever setup that matches the description.

 1	 2	 3

Feature	Lever set up number
Effort arm is longer than the load arm	
Load arm is longer than the effort arm	
Load arm is equal to the effort arm	
Lifted the book with the greatest ease (provided the greatest mechanical advantage)	

Classes of Lever

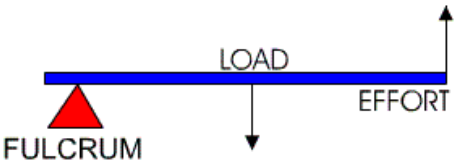
CLASS ONE

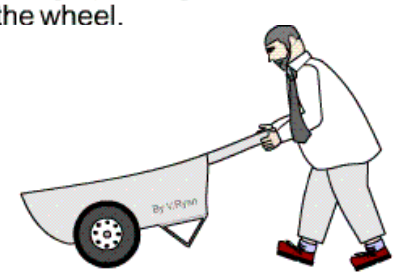




The workman uses a trolley to move the large packing case. The fulcrum is the wheel.

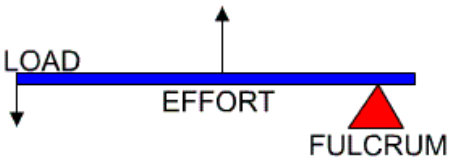
CLASS TWO

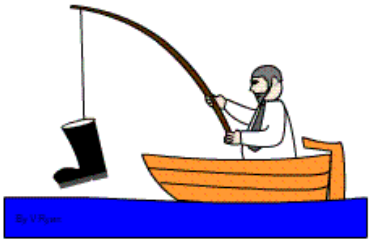




The gardener uses a wheel barrow to lift tools and garden waste. The load is in the centre of the barrow

CLASS THREE





The fisherman catches the 'fish' which becomes the load at the end of the lever.

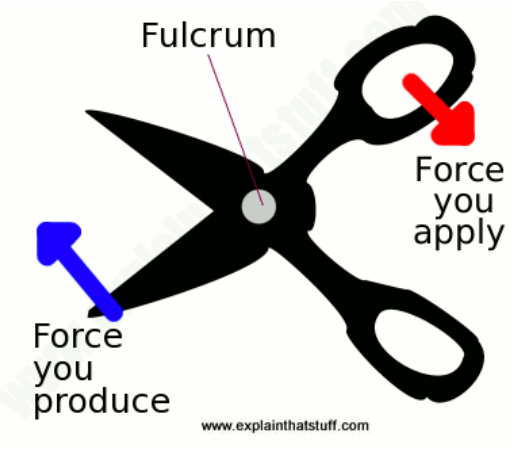
Whether a lever is class 1, class 2 or class 3 depends on the order of the effort, load and fulcrum.

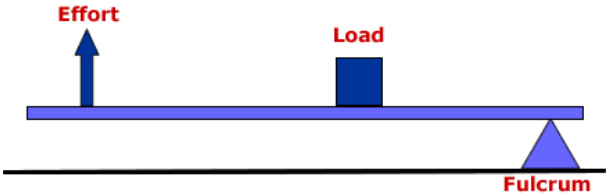
Lever Class	Which is in the middle? (effort, load or fulcrum)
1	
2	
3	

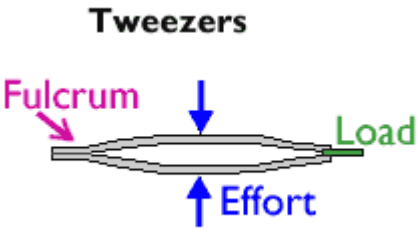
1. Why would remember the acronym "F.L.E" help to identify which class of lever is which?

2. Write down whether each lever is a class 1, 2 or 3.









3. Label the effort, load and fulcrum AND write down if the lever is class 1, 2 or 3.





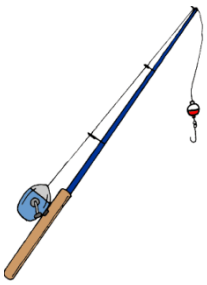
Levers provide a Mechanical Advantage (MA)

There are two types of mechanical advantages-

1. Force advantage - this type of lever enables a big load to be lifted with a small effort. These levers have an MA greater than one.



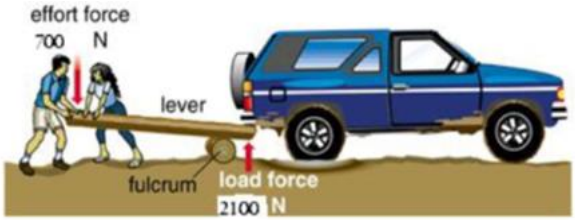
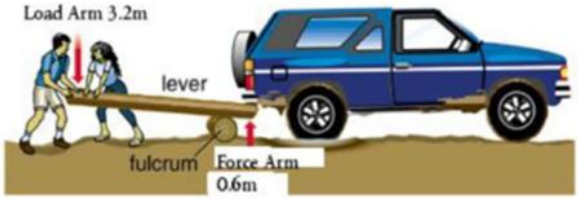
2. Speed/distance advantage - this type of lever enables a load to be moved faster or over a greater distance with no greater effort. These levers have a MA less than one.



Mechanical advantages can be calculated by using the following formula.

$$MA = \frac{Load\ Force}{Effort\ Force} \qquad Or \qquad MA = \frac{length\ of\ effort\ arm}{length\ of\ Load\ arm}.$$

Identify which formula would be used to work pout the MA in each of the following situations.

Situation	 Situation 1	 Situation 2
Formula		

Carryout the following investigation on the effect of increasing the effort arm on the mechanical advantage.

Effort (g) 100g = 1N	Load (g)	Effort Arm (cm)	Load arm (cm)	Relationship Between Effort and load arm	MA = load / effort	Type of mechanical advantage. Force advantage (MA = greater than 1) Speed distance advantage (MA= less than 1) MA of 0 = no advantage
	100	25	25	Effort arm equal to load arm		
	100	30	20	Effort arm is		
	100	35	15	Effort arm is		

Complete the following summary by deleting the incorrect response.

1. When the effort arm is longer than the load arm an MA greater / less than one is achieved.
2. An MA greater than one provides a force / speed, distance advantage.
3. As the effort arm increases the MA increases / decreases.

Investigate the effect of increasing the load arm on the mechanical advantage.

Effort (g) 100g = 1N	Load (g)	Effort Arm (cm)	Load arm (cm)	Relationship Between Effort and load arm	MA = load / effort	Type of mechanical advantage Force advantage (Ma = greater than 1) Speed distance advantage (MA= less than 1)
	100	25	25	Load arm equal to effort arm		
	100	20	30	Load arm is		
	100	15	35	Load arm is		

Complete the following summary by deleting the incorrect response.

1. When the load arm is longer than the effort arm an MA greater / less than one is achieved.
2. An MA less than one provides a force / speed, distance advantage.
3. As the load arm increases the MA increases / decreases.

Lever Summary

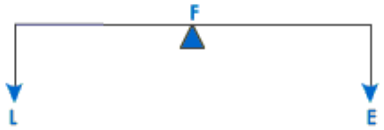
Length of effort and load arms	Type of Advantage	MA	Diagram
Effort arm = load arm	No force advantage No distance advantage Changes the direction the force is acting	1	
Effort arm greater than the load arm	Provides a Force advantage. A bigger load can be raised with a smaller effort	Greater than 1	
Load arm greater than the effort arm	Provides a speed/distance advantage The load is moved a long way . The effort needed is greater than the load.	Less than 1	

Figure 8.1.8 shows how this applies to opening a tin.

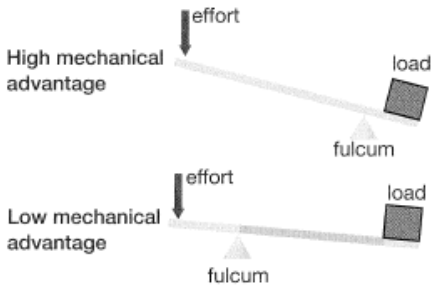


Figure 8.1.7 As the distance between the effort and the fulcrum increases, so too does the mechanical advantage of a lever.

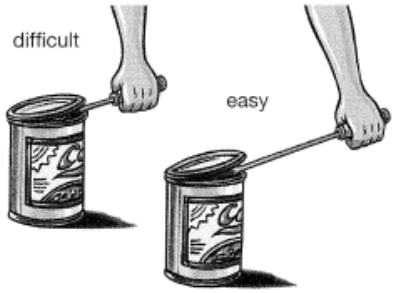


Figure 8.1.8 It is easier to open the tin shown using the longer screwdriver, because it has a greater distance from the fulcrum to the effort.

Problem 1

Jess applies an effort of 150 N to lift a rock of weight 600 N using a long piece of wood as a lever. Calculate the mechanical advantage of the lever.

Solution 1

The load is the weight Jess is trying to lift. This is 600 N.

The effort is the force she exerts of 150 N.

$$\text{Mechanical advantage} = \frac{\text{load}}{\text{effort}}$$

$$= \frac{600}{150}$$

$$= 4$$

This means that the lever has made Jess's force four times larger. She only needs to put in one-quarter the effort she would otherwise need to apply to do the job.

Problem 2

Con uses another stick to shift a second rock. The distance from his hand on the stick to the fulcrum is 120 cm, and the distance from the fulcrum to the rock is 40 cm. Calculate the mechanical advantage of this lever system.

Solution 2

$$\text{Mechanical advantage} = \frac{\text{distance from effort to fulcrum}}{\text{distance from load to fulcrum}}$$

$$= \frac{120}{40}$$

$$= 3$$

This means Con can shift the rock with three times less force than when using his bare hands. It also means that he could lift a rock three times heavier than he normally could.

Problem Solving with Levers - apply your knowledge.

Set 1: Calculate the MA values for the following levers:

a. **L** **E**

10cm ^ **25cm**

Values from the diagram Effort arm = Load arm =	
Formula you will use	
Substitute the values into the formula	
Answer	

b. **L = 220N** **E = 65N**

^

Values from the diagram Effort = Load =	
Formula you will use	
Substitute the values into the formula	
Answer	

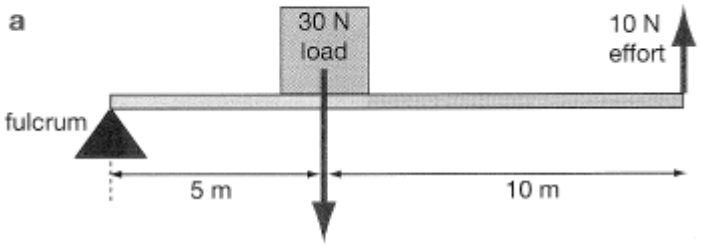
c. **L** **E**



Values from the diagram Effort arm = Load arm =	
Formula you will use	
Substitute the values into the formula	
Answer	

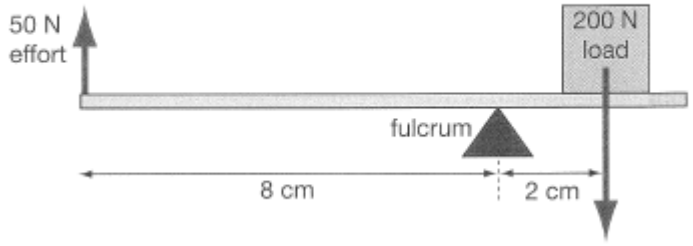
Set 2: Calculate the MA values for the following levers:

12 **Calculate** the size of the mechanical advantage for each system shown in Figure 8.1.12.



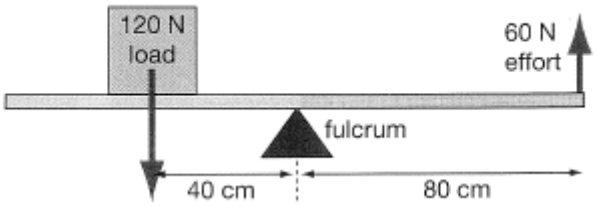
Values from the diagram	
Formula you will use	
Substitute the values into the formula	
Answer	

b



Values from the diagram	
Formula you will use	
Substitute the values into the formula	
Answer	

c



Values from the diagram	
Formula you will use	
Substitute the values into the formula	
Answer (don't forget the units)	

Set 3: Calculate the MA values for the following levers

1 You apply a force of 20 N to prise the lid off a Milo tin using a teaspoon. In doing so, you needed to overcome a force of 120 N. **Calculate** the mechanical advantage of the teaspoon lever.

2 Sam needs to remove an old wooden sleeper when designing a garden bed. He uses a crowbar, with a small rock for the fulcrum. The crowbar is 2 metres long, and the small rock sits 50 cm from the sleeper.

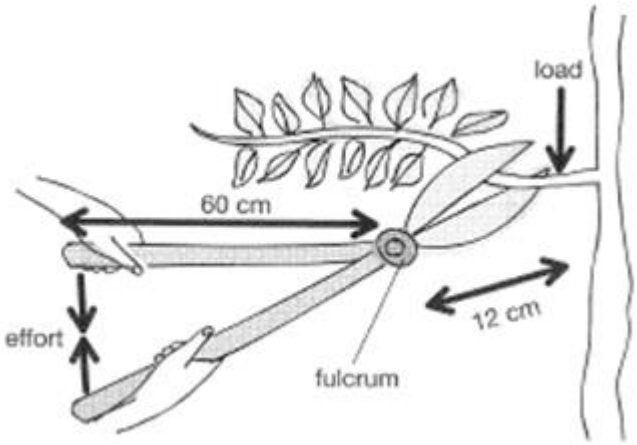
(a) Construct a diagram of this situation.

(b) State the distance from the load to the fulcrum.

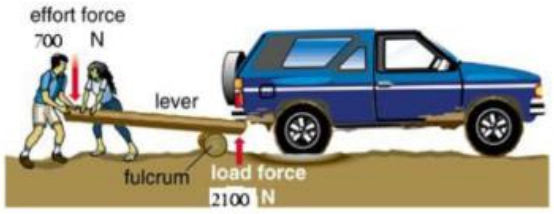
(c) State the distance from the effort to the fulcrum.

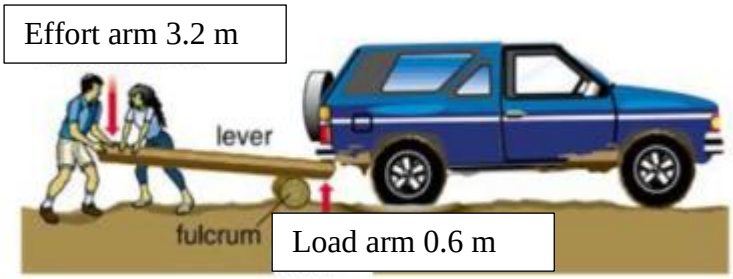
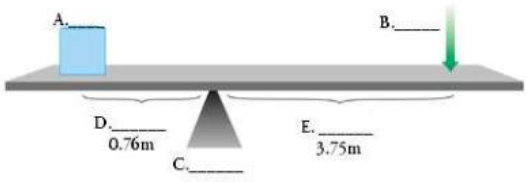
(d) Calculate the mechanical advantage.

3. Calculate the mechanical advantage provided by the garden clippers. Show working.



Set 4: Calculating mechanical advantage.

Question 1	 Susan and Jake's truck gets stuck in the mud. They used a tree branch as a lever to lift the truck out of the mud. They applied an effort force of 700N to the branch and the back of the truck weighs 2100N. What is the mechanical advantage of the tree branch lever?
Answer Show working	
Question 2	Susan and Jake take Susan's car for a drive and get suck in the mud. The back of Susan's car weighs 900N. They applied 700N of force to the tree branch lever. What is the mechanical advantage of the tree branch?
Answer Show working	
Question 3	Marc is asked to raise a flag at the Remembrance Day ceremonies at his school. If the effort force to raise the flag is 50N and the load force—the flag plus the rope—is 50N what is the mechanical advantage of the pulley on the flagpole?
Answer Show working	
Question 4	Justin is riding his bike up Dunmore Hill. The effort force from his feet on his pedals is 1160N. The resulting load force to move the bike forward is 675N. What is the mechanical advantage of the bike?
Answer Show working	

Question 5	 **What is the MA of the branch Susan and Jake are using?**
Answer Show working	
Question 6	If Susan and Jake made the force arm 0.2m instead of 3.2m then what is the mechanical advantage? Is this easier or harder to lift the load.
Answer Show working State units	
Question 7	 With the information given calculate the mechanical advantage.
Answer Show working	

Test Your Knowledge - highlight the correct response.

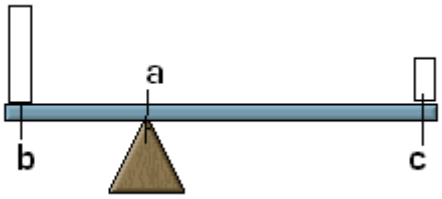


Figure 1

1. Figure 1 is an example of:
a) Class 1 lever
b) Class 2 lever
c) Class 3 lever
d) Class 4 lever

2. Which of the following statement is true for Figure 1.
a) b is the fulcrum, c is the load, a is the effort
b) b is the load, c is the fulcrum, a is the effort
c)b is the fulcrum, a is the effort, c is the load.
d) b is the load, a is the fulcrum, c is the effort

3. In figure 1, if the distance from a to b is 20cm, and the distance from a to c is 80 cm, then the mechanical advantage of the system is.
a)20
b) 80
c)4
d) 1/4

4. Given that the mass at (b) is 1 kilogram. What mass would the block at (c) have to be to lift block (b)? Note: Use the distance values given in question 3.
a) 1000 grams
b) 250 grams
c) 80 grams
d) 400 grams

5. If the mechanical advantage of a simple machine is 4, then the
a) output force is 4 times the effort
b) effort is 4 times the output force
c) efficiency is 4%
d) the work output is 4 times the input